

Simplex This Far

$$\min: \vec{c}^T \vec{x}$$

$$\text{st: } A\vec{x} = \vec{b}, \quad A \text{ } m \times n$$

$$\vec{x} \geq 0$$

$$B \subseteq [n], \quad |B| = m$$

$$\vec{x} = \begin{pmatrix} \vec{x}_B \\ 0 \end{pmatrix}, \quad \vec{x}_B = A_B^{-1} \vec{b} \quad |$$

$$\underline{x_i = 0, \quad \forall i \notin B}$$

Move to adjacent BFS by choosing
new B_{new} , $|B \cap B_{\text{new}}| = m-1$

$$B_{\text{new}} = (B \setminus \hat{j}) \cup \hat{i}$$

$$\vec{x}_{\text{new}} = \vec{x} + \theta \vec{d} \quad \geq 0$$

$$\vec{d} = \begin{pmatrix} d_B \\ 1 \\ 0 \end{pmatrix} \leftarrow \hat{i} \quad //$$

$$A(\vec{x} + \theta \vec{d}) = \vec{b}$$

$$\Rightarrow A\vec{d} = 0$$

$$\Rightarrow \vec{d}_B = -A_B^{-1} A_{\hat{i}}$$

$$\theta = \min_{j: d_j > 0} \frac{x_j}{d_j}$$

Reduced Costs

Did we improve? Is $C^T x_{\text{new}} < C^T x$

$$x_{\text{new}} = \begin{pmatrix} x_B \\ 0 \end{pmatrix} - \theta \begin{pmatrix} A_B^{-T} A_i^T \\ 0 \end{pmatrix} + \theta e_i$$

$$C^T x_{\text{new}} = \underbrace{C_B^T x_B} - \theta C_B^T A_B^{-T} A_i^T + \theta C_i$$

$$\underbrace{C^T x_{\text{new}} - C^T x} = \theta \underbrace{\left(C_i - (C_B^T A_B^{-T}) A_i^T \right)}$$

Reduced cost of $\hat{i}$.

$$\boxed{\bar{C}_i \triangleq C_i - (C_B^T A_B^{-T}) A_i^T}$$

• If $i \in B$: $\bar{C}_i = 0$

• If $\bar{C}_i < 0$, then we might improve by adding i to basis if $\theta \neq 0$

• If $\bar{C} \geq 0 \Rightarrow x$ is an optimal solution.

Summary of Conceptual Simplex Method

$$\text{At } x = \begin{pmatrix} x_B \\ 0 \end{pmatrix}$$

$$x_B = A_B^{-1} b$$

$$\bar{c} = c - c_B A_B^{-1} A$$

If $\bar{c} \geq 0 \Rightarrow x$ optimal.

Else: pick $\hat{i} : c_{\hat{i}} < 0$

$$\text{compute: } d_B = -A_B^{-1} A_{\hat{i}}, \quad d_{\hat{i}} = 1.$$

$$\theta = \min_{j: d_j > 0} \frac{x_j}{d_j}$$

// gives new B

$$x_{\text{new}} = x + \theta d = \begin{pmatrix} x_{B_{\text{new}}} \\ 0 \end{pmatrix}, \quad x_{B_{\text{new}}} = A_{B_{\text{new}}}^{-1} b.$$

Discussion: Computational Considerations

A. Naive impl.

Could potentially cycle forever,
if the problem degenerate.

B. Simple fix guarantees that simplex
terminates in finite time

Could take exp'd time,
(smoothed analysis)

C. A_B^{-1}